

Data Structures – Unit 10

Bucharest University of Economic Studies

Binary Trees



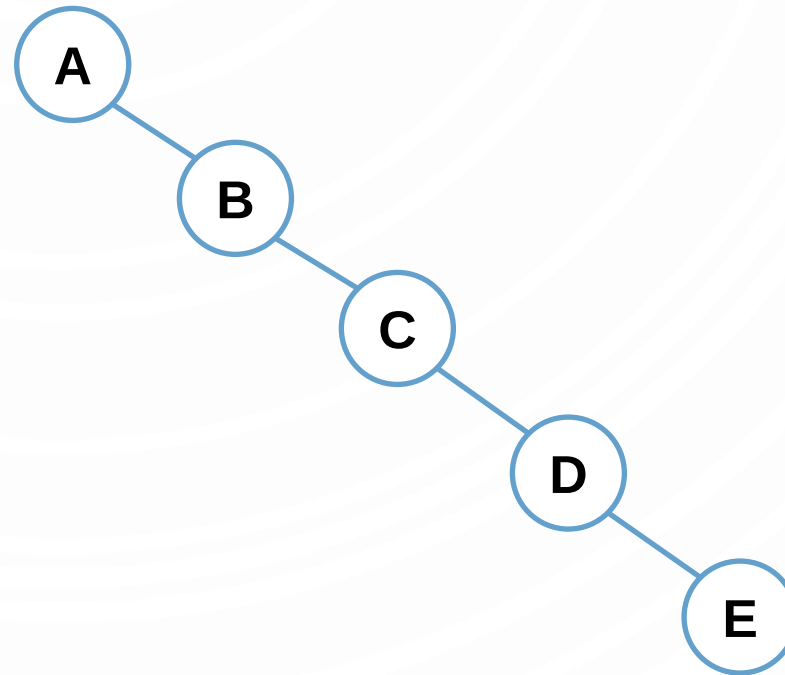
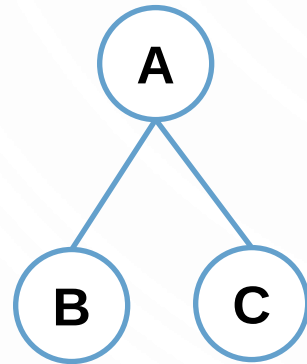
Binary Trees - Characteristics

- A generic tree with maximum 2 descendants;
- Any subtree has the same restriction mentioned above;
- Binary tree topologies: binary search trees, expression trees;
- Representation: HEAP memorized structure.



Binary Trees

Each node has maximum 2 descendants:



Expression binary trees

- Each leaf is an operand;
- Each non-leaf node is an operator;

```
struct expTreeNode
```

```
{
```

```
    int key;
```

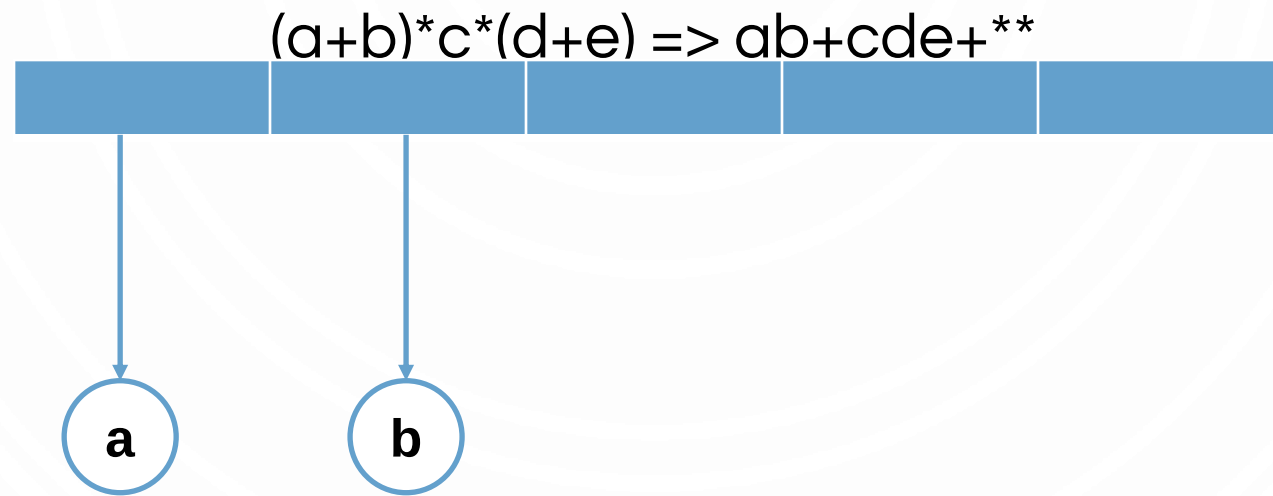
```
    expTreeNode *left;
```

```
    expTreeNode *right;
```

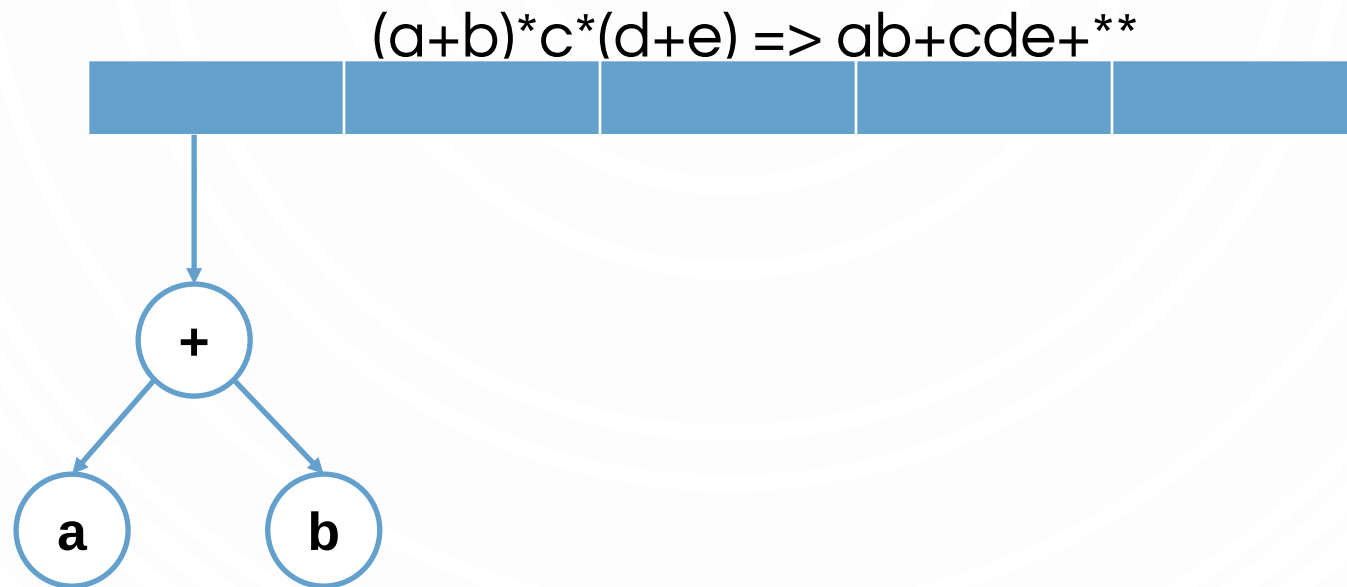
```
}
```



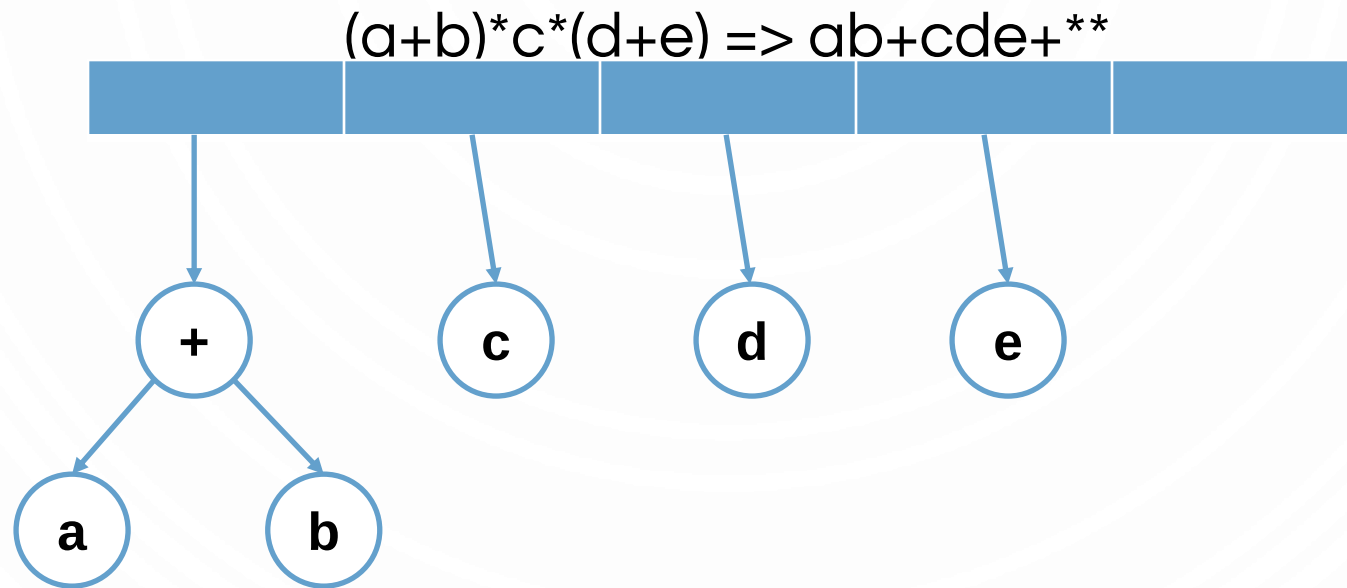
Expression binary trees



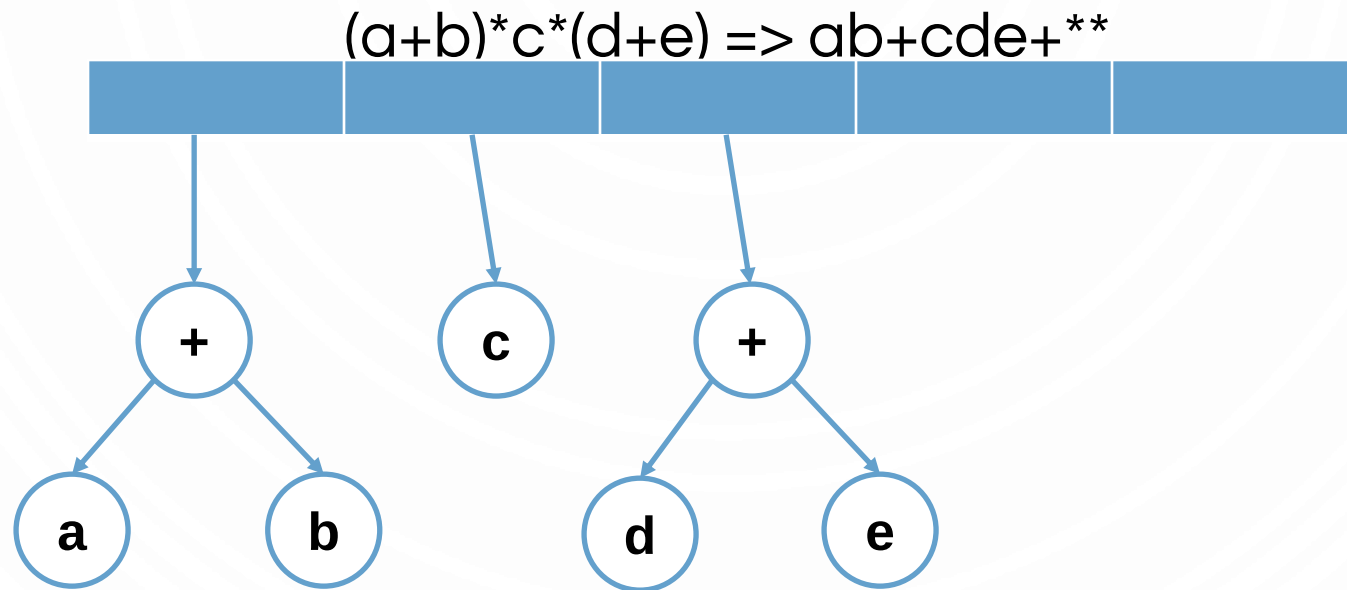
Expression binary trees



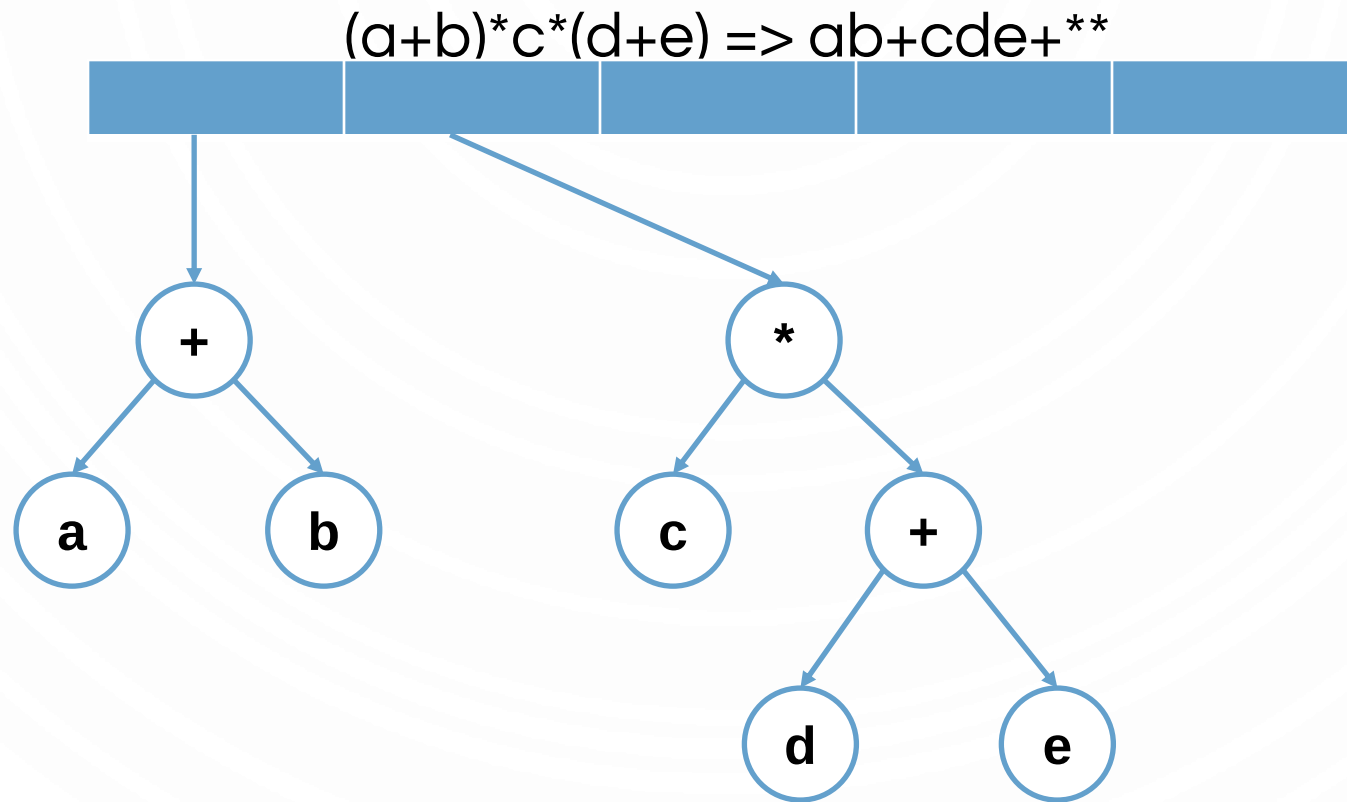
Expression binary trees



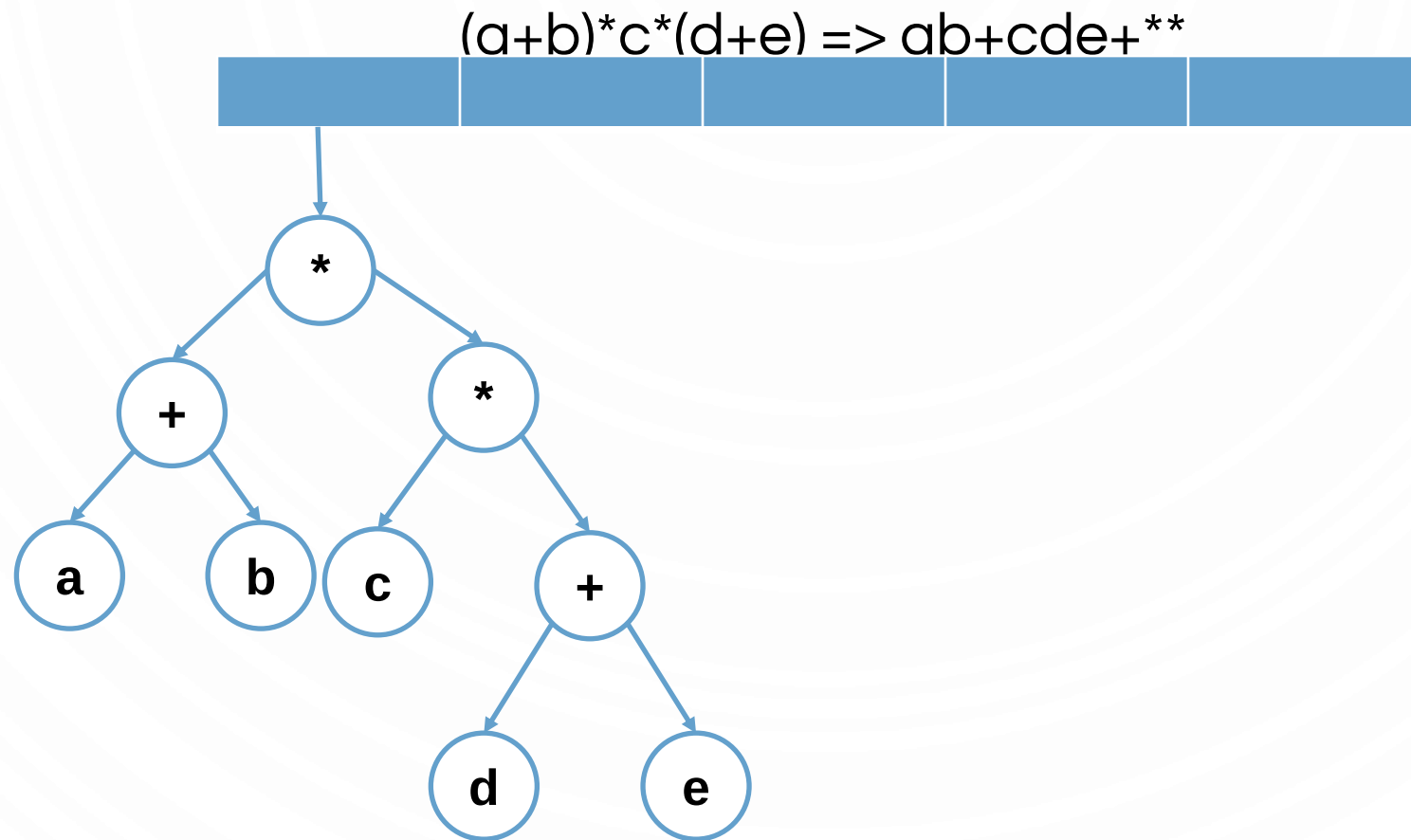
Expression binary trees



Expression binary trees



Expression binary trees

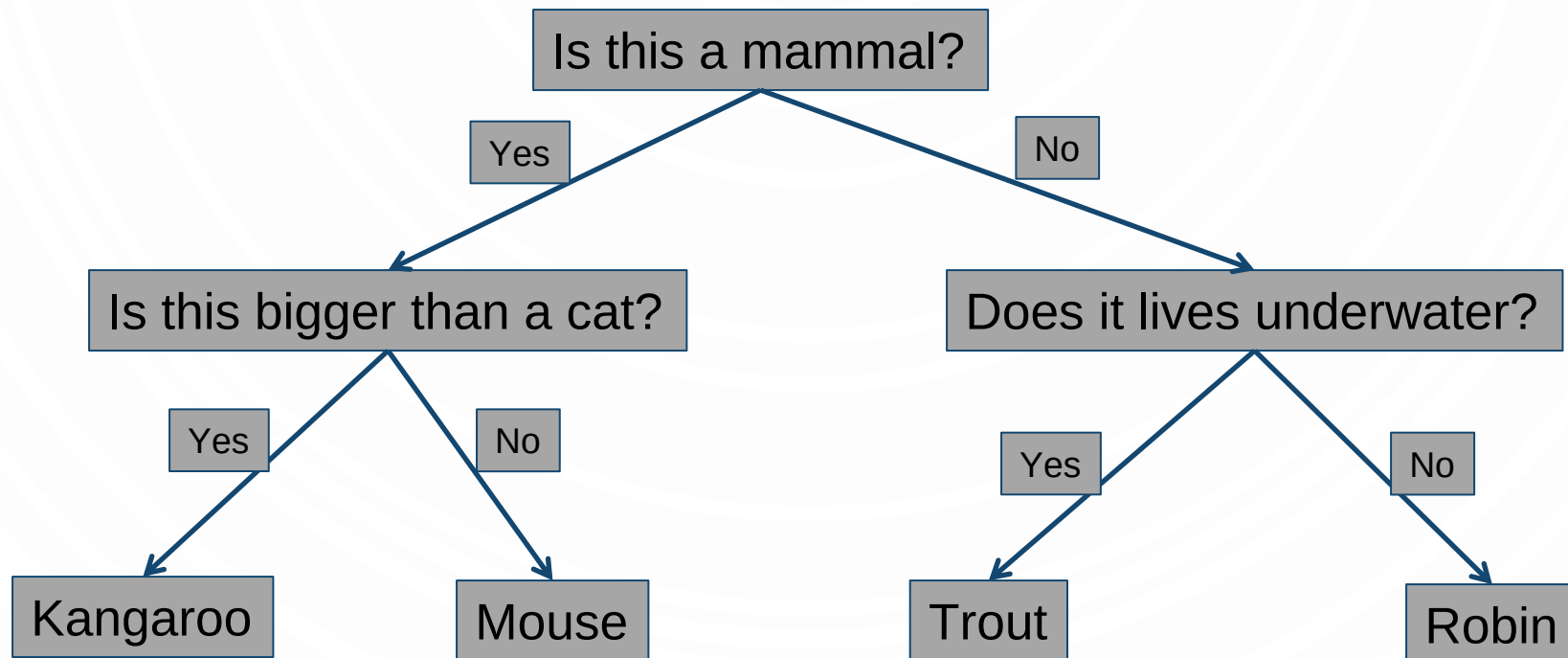


Taxonomy trees

- Decision trees;
- Trees used to store knowledge;
- Each leaf contains an object;
- Each non-leaf node contains a decision about the objects below it;



Taxonomy binary trees



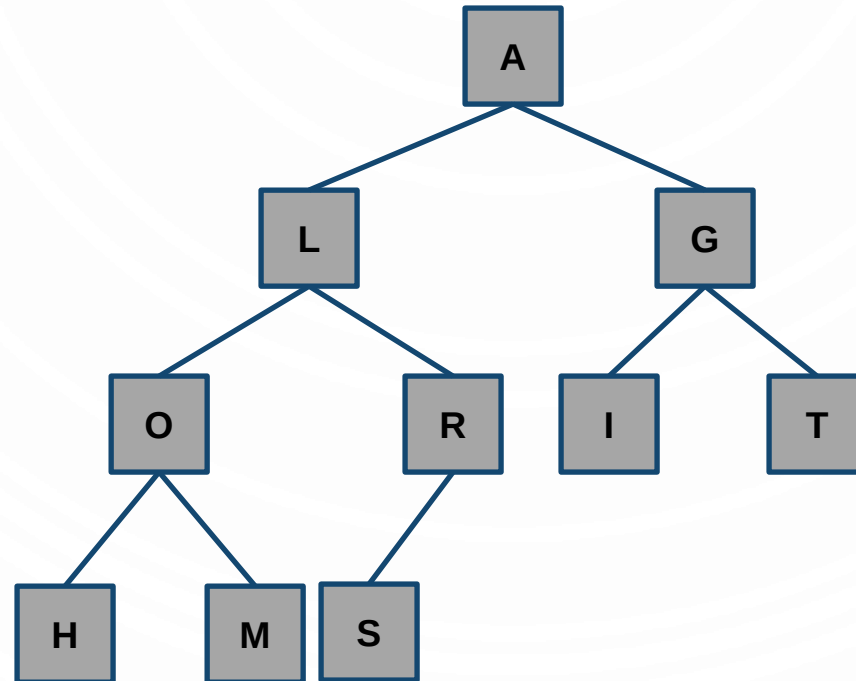
Full vs. Complete BTs

A full binary tree is a structure in which each node excepting the leaves have exactly two children;

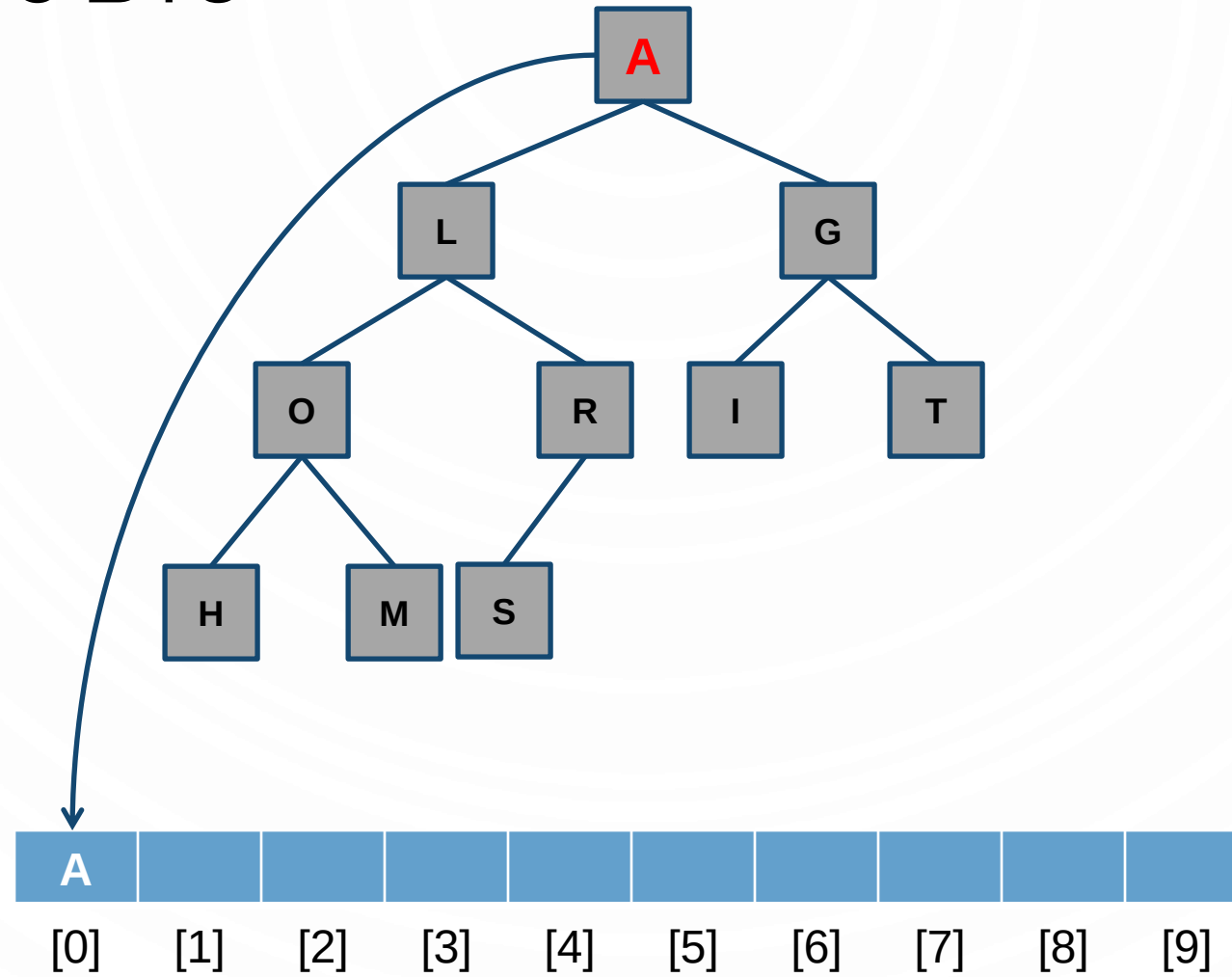
A complete binary tree is a tree in which every level, except possibly the last, is completely filled, and all nodes are as far left as possible;



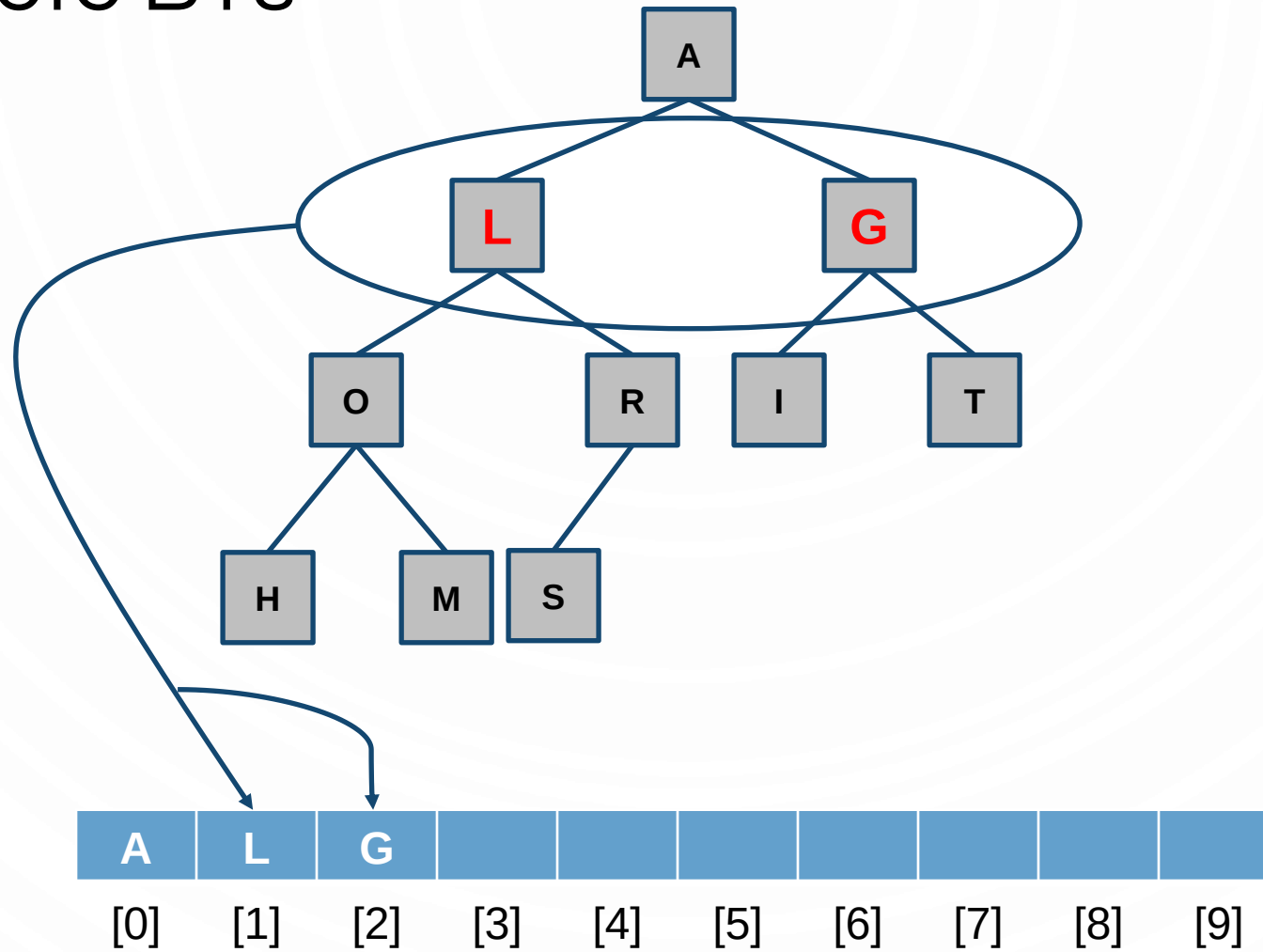
Complete BTs



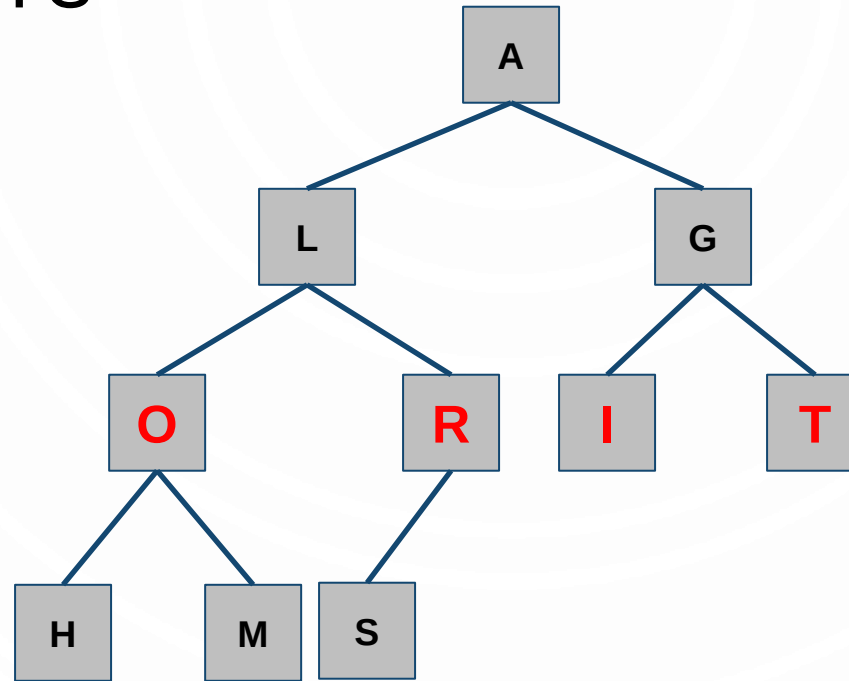
Complete BTs



Complete BTs



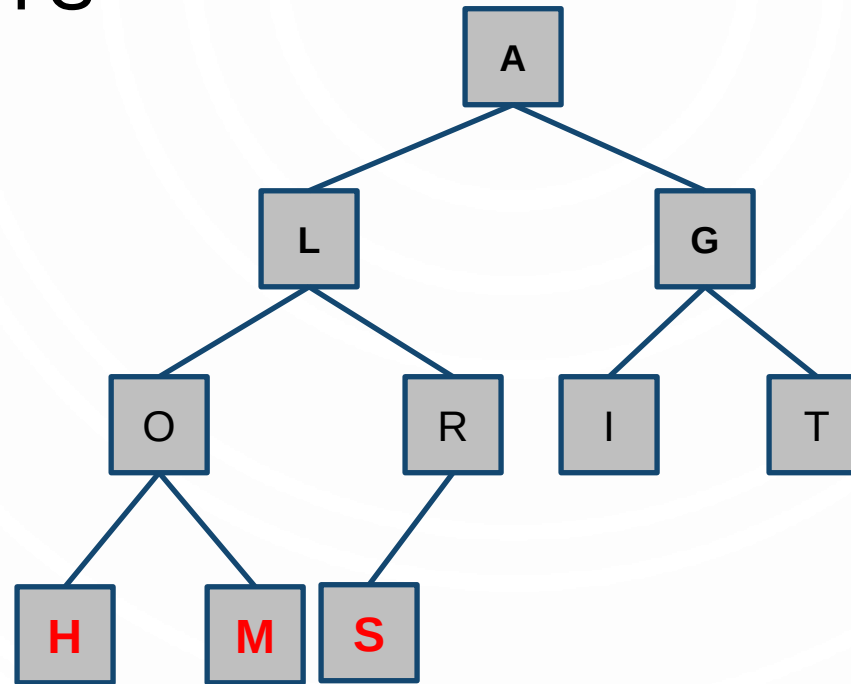
Complete BTs



A	L	G	O	R	I	T			
[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]



Complete BTs

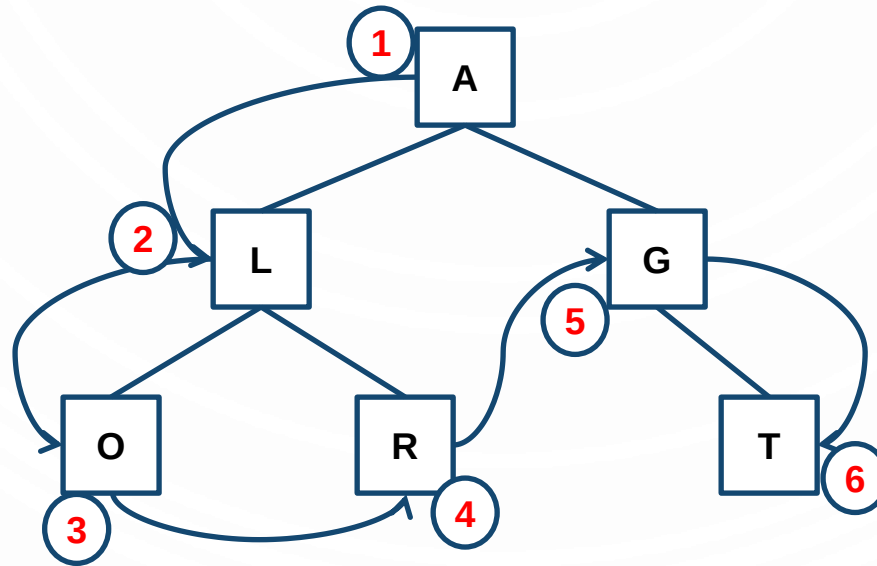


A	L	G	O	R	I	T	H	M	S
[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]



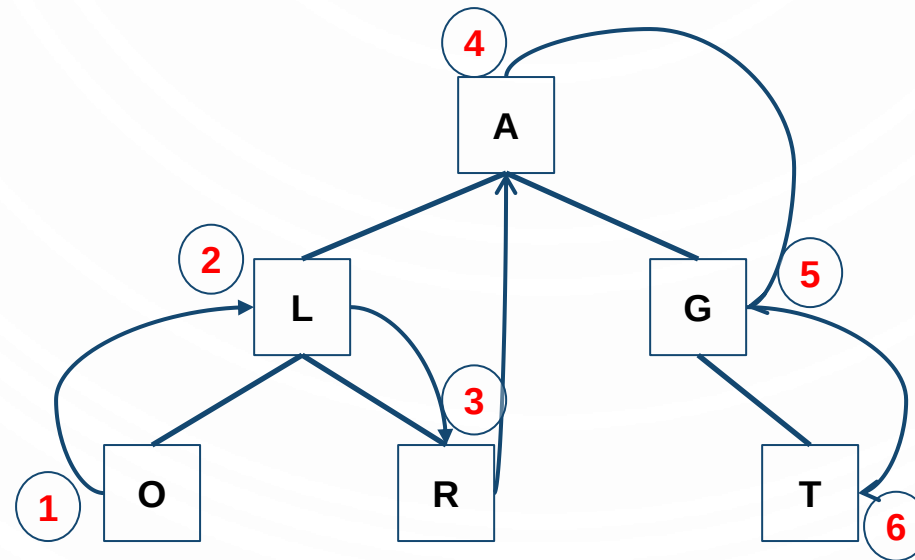
Binary tree traversal

Pre-order: root / left node / right node;



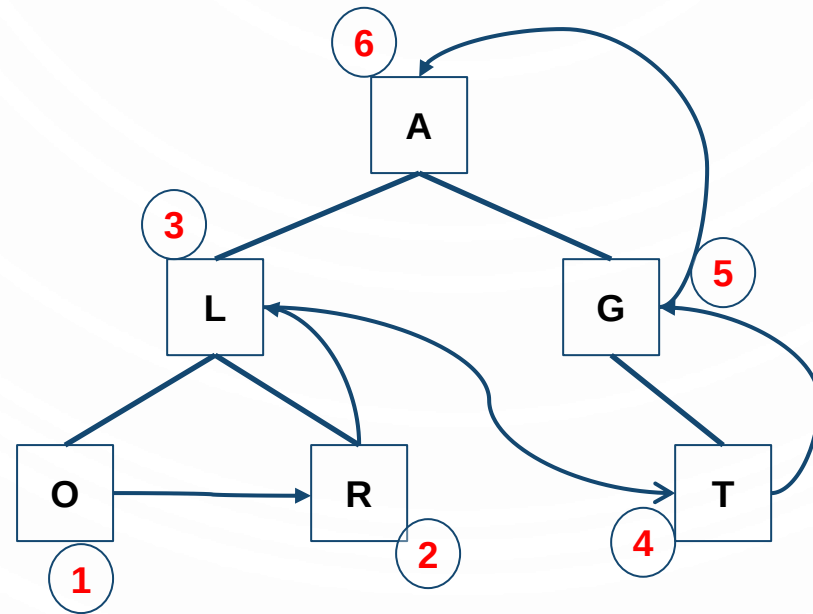
Binary tree traversal

In-order: left node / root / right node;



Binary tree traversal

Post-order: left node / right node / root ;



Complexity

The complexity for a binary search tree operation is:

$$O(\log n),$$

where n represents the number of elements in the array.



BST - Definition

A binary tree in which each node follows and implements a total order semantics is considered a **B**inary **S**earch **T**ree.



Binary Search Trees

- Total order semantic:

- $==, !=, >=, <=, >, <$

- Equality

- Completeness

- Consistency

- Transitivity



Binary Search Trees

Equality

- for x, y , values of two different nodes $(x==y) \Leftrightarrow x$ and y are identical;



Binary Search Trees

Completeness

- For any x, y , values of two different nodes, one of the following relations is unique and true:
 - $(x < y)$;
 - $(x == y)$;
 - $(x > y)$;



Binary Search Trees

Consistency

- For any x, y , values of two different nodes, the relations:
 - $(x > y)$ is identical with $(y < x)$;
 - $(x \geq y)$ is identical with $(y \leq x)$ identical with $((x > y) \parallel (x == y))$;
 - $(x \leq y)$ is identical with $(y \geq x)$ identical with $((x < y) \parallel (x == y))$;
 - $(x \neq y)$ is identical with $!(x == y)$;



Binary Search Trees

Transitivity

- for any x, y, z , values of three different nodes, if $(x < y)$ and $(y < z)$ then $(x < z)$ is true;

Binary Search Trees

Rules for storing elements in a BST:

R1: in BST elements are compared between them using the total order semantic;

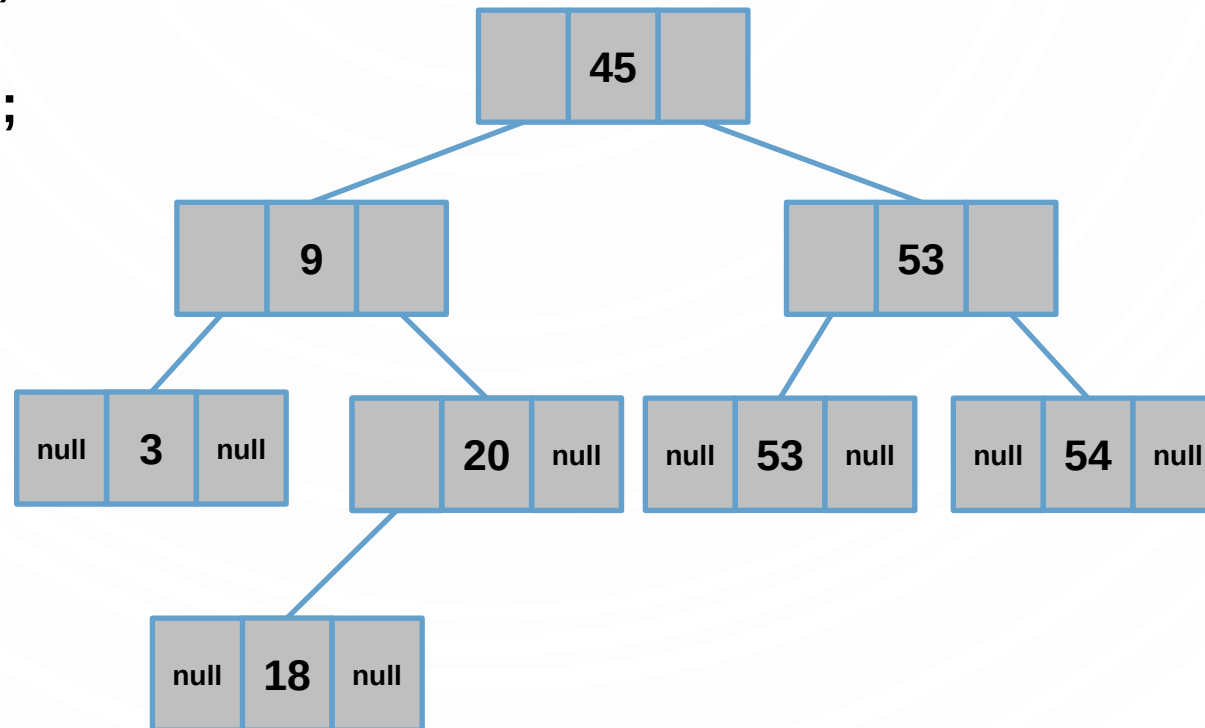
R2: the following rules apply to each node:

- each element from the left subtree of the current node is smaller than the current element;
- each element from the right subtree of the current node is larger than the current element;



BST - Representation

```
struct BSTNode{  
    BSTNode * ls;  
    int info;  
    BSTNode * rs;  
};
```



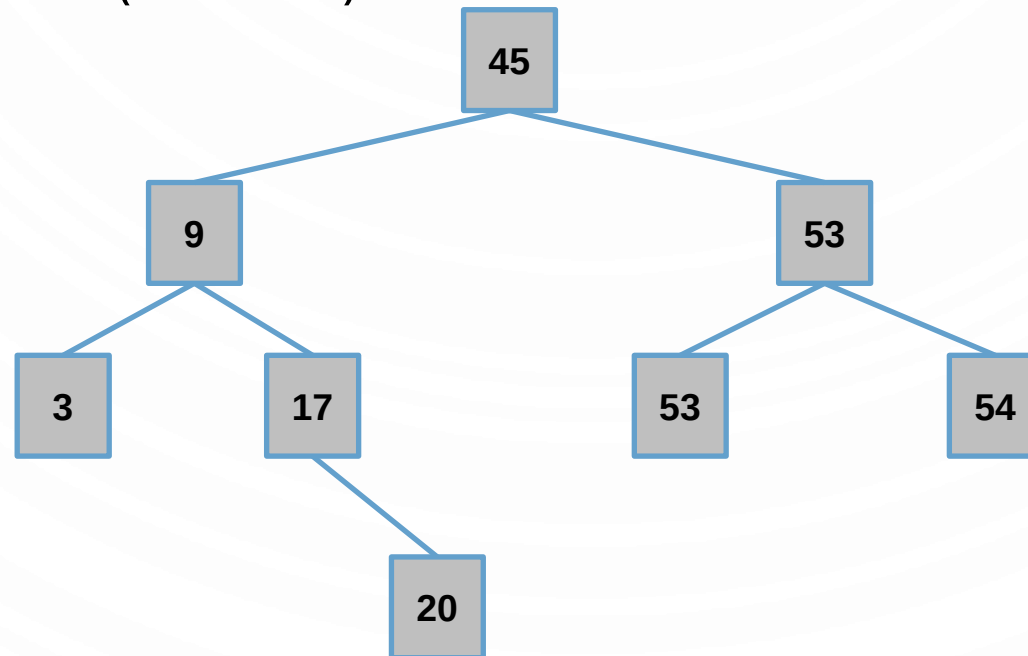
BST - Operations

- Structure traversal;
- Search for min and max;
- Insertion operations;
- Deletion with keeping the total order semantic;
- Calculating the height of a tree;
- Concatenating two trees;

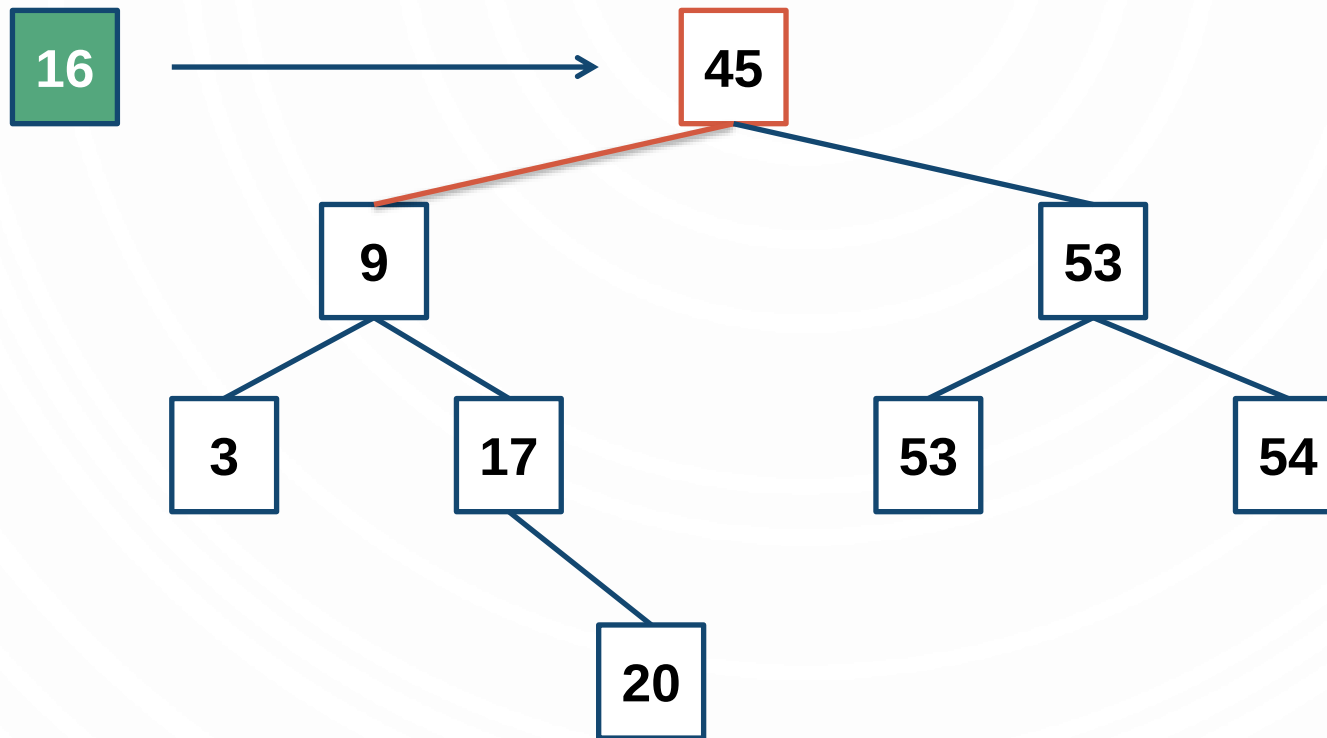


BST - Traversal

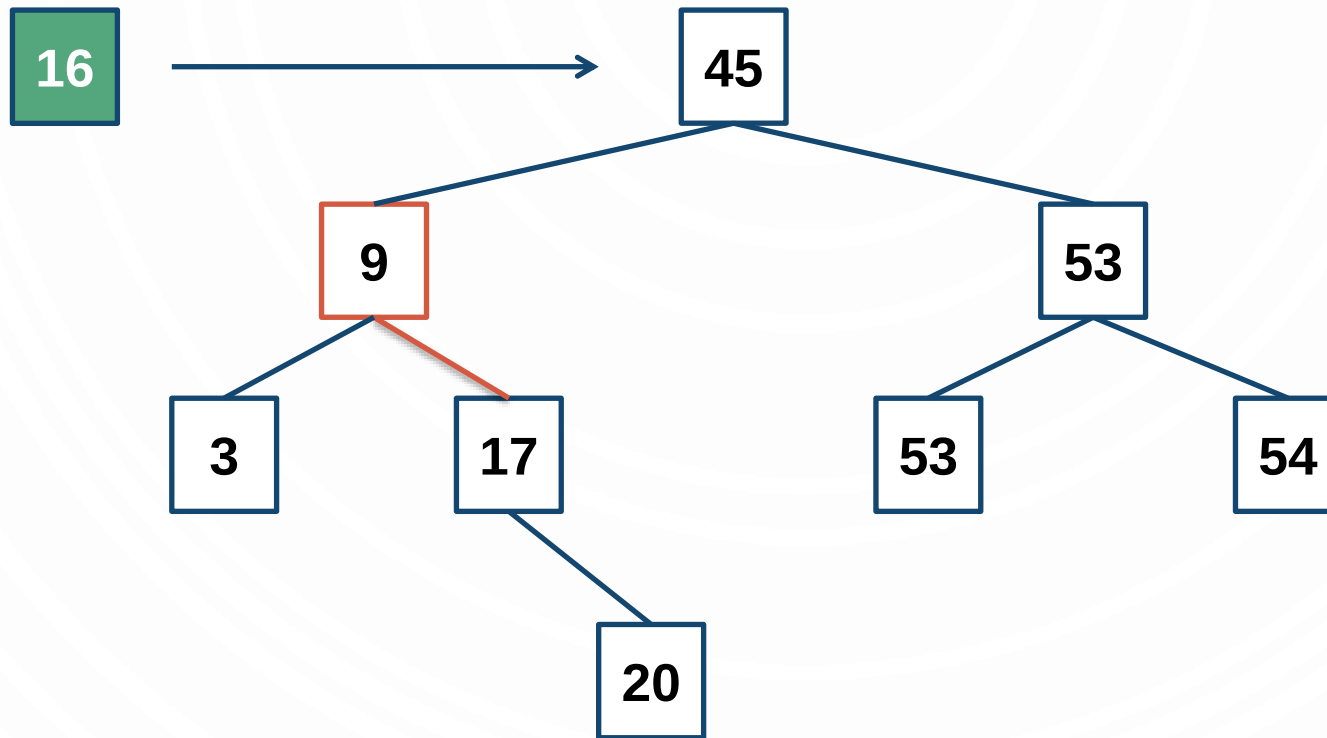
- Post-order (RootLR): 45 / 9 / 3 / 17 / 20 / 53 / 53 / 54
- In-order (LRootR): 3 / 9 / 17 / 20 / 45 / 53 / 53 / 54
- Pre-order (LRRoot): 3 / 20 / 17 / 9 / 53 / 54 / 53 / 45



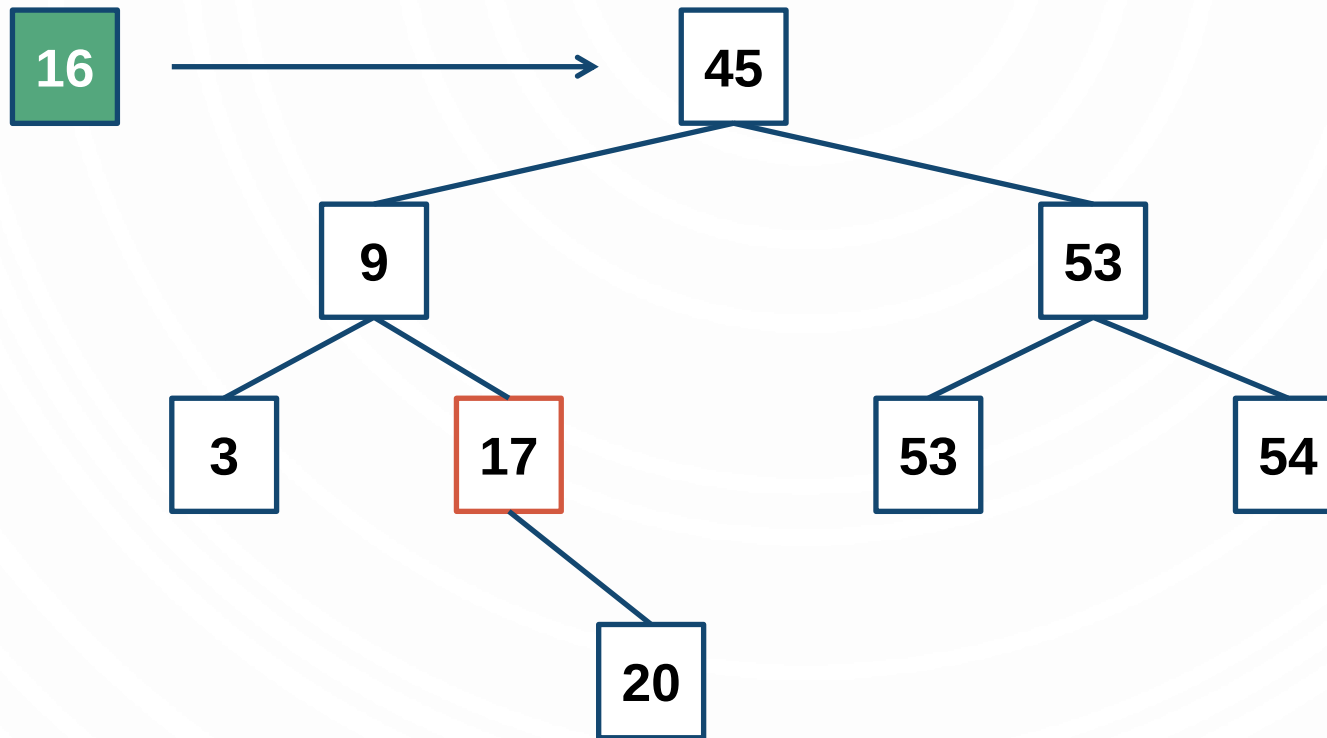
BST – Insert operation



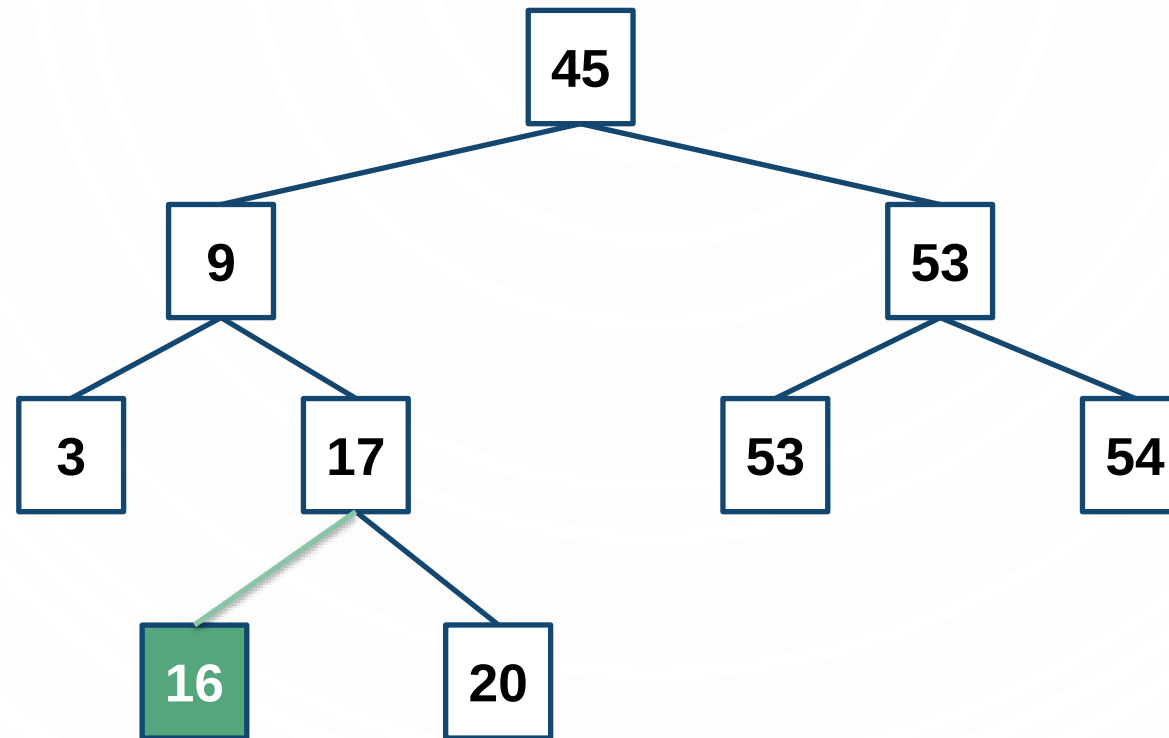
BST – Insert operation



BST – Insert operation



BST – Insert operation



BST – Insert operation

```
template<class T> struct Student{
```

```
    T recordId;
```

```
    T grade;
```

```
};
```

```
template<class T> struct TreeNode{
```

```
    T* info;
```

```
    TreeNode<T>* ss;
```

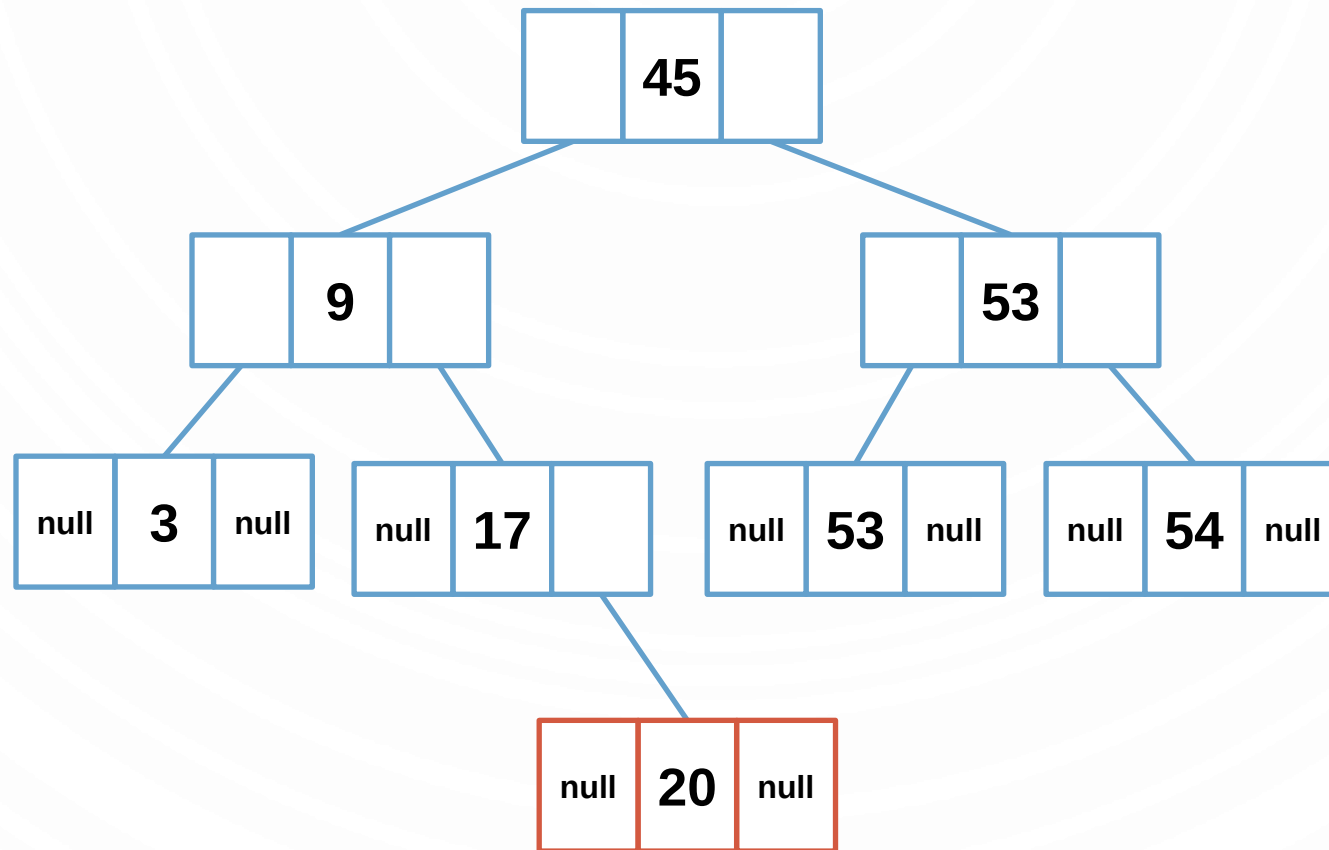
```
    TreeNode<T>* sd;
```

```
};
```



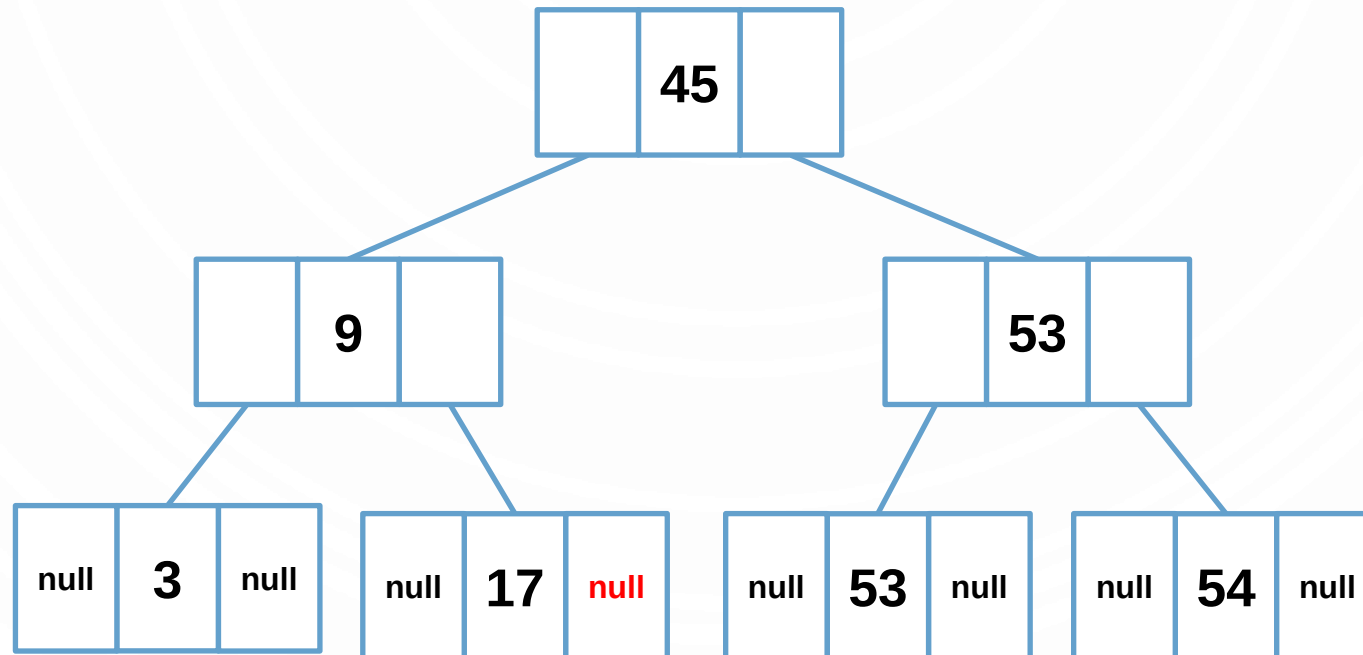
BST – Delete operation

Case 1: deletion of a leaf node **20**



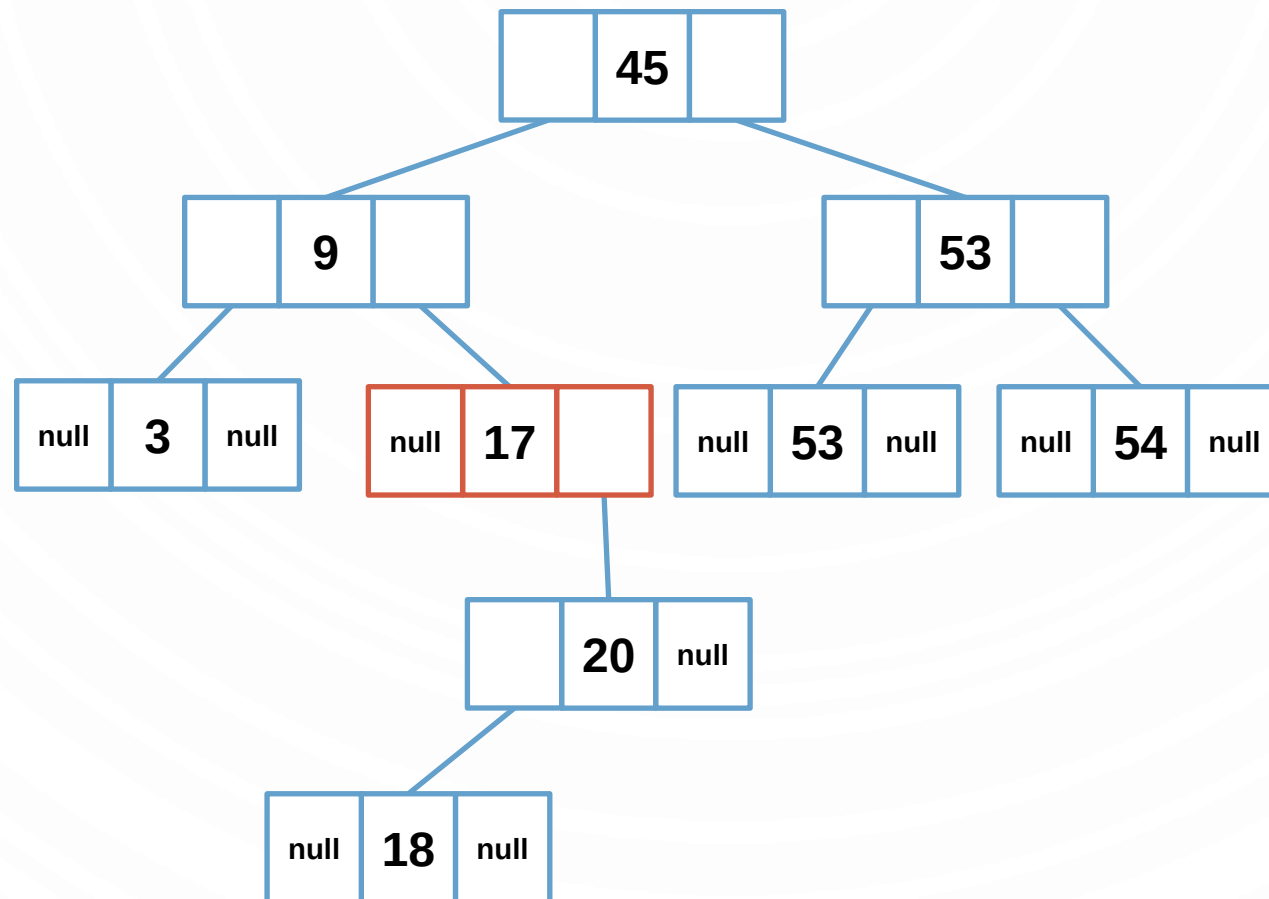
BST – Delete operation

Case 1: deletion of a leaf node **20**



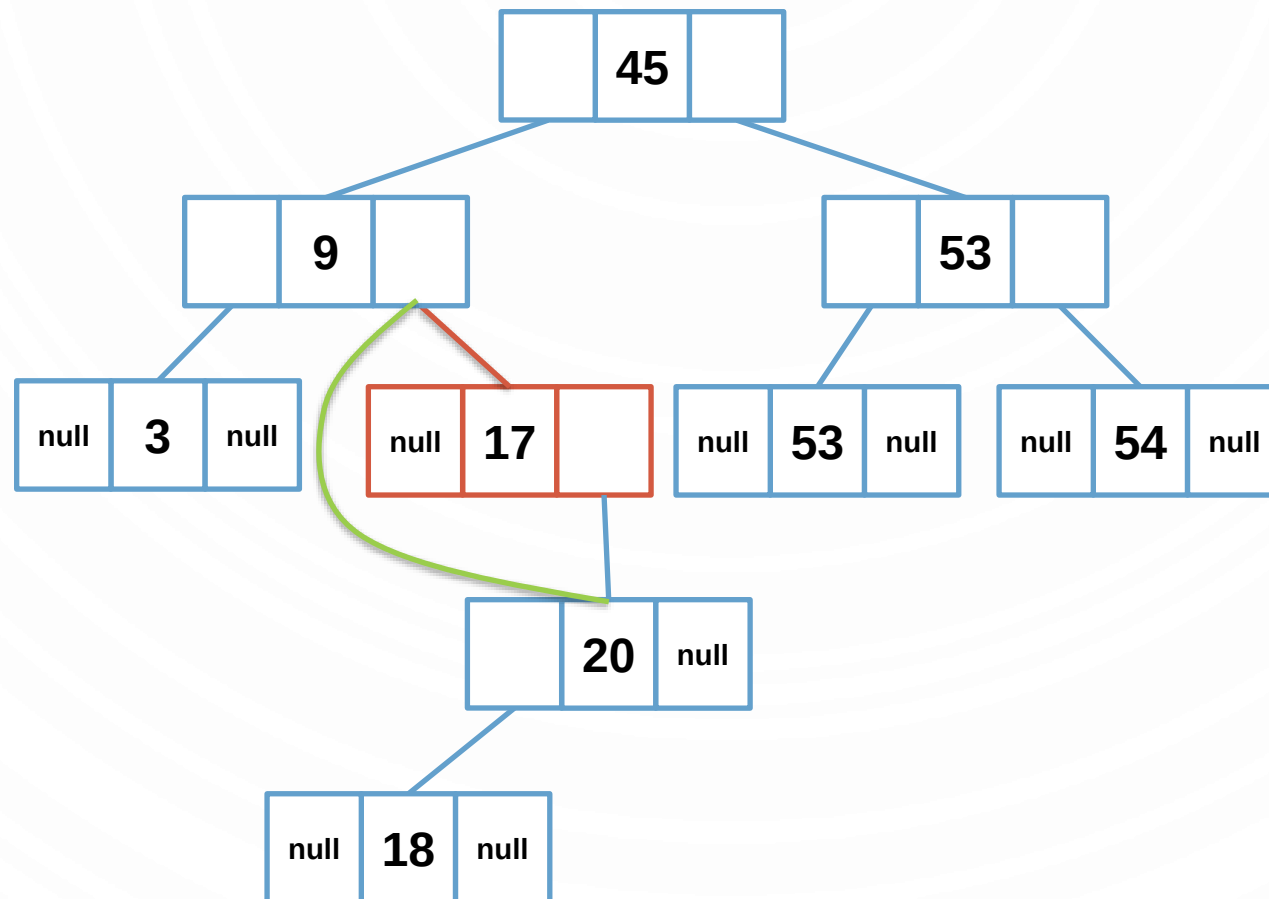
BST – Delete operation

Case 2: deletion of a node with only the right descendant, **17**



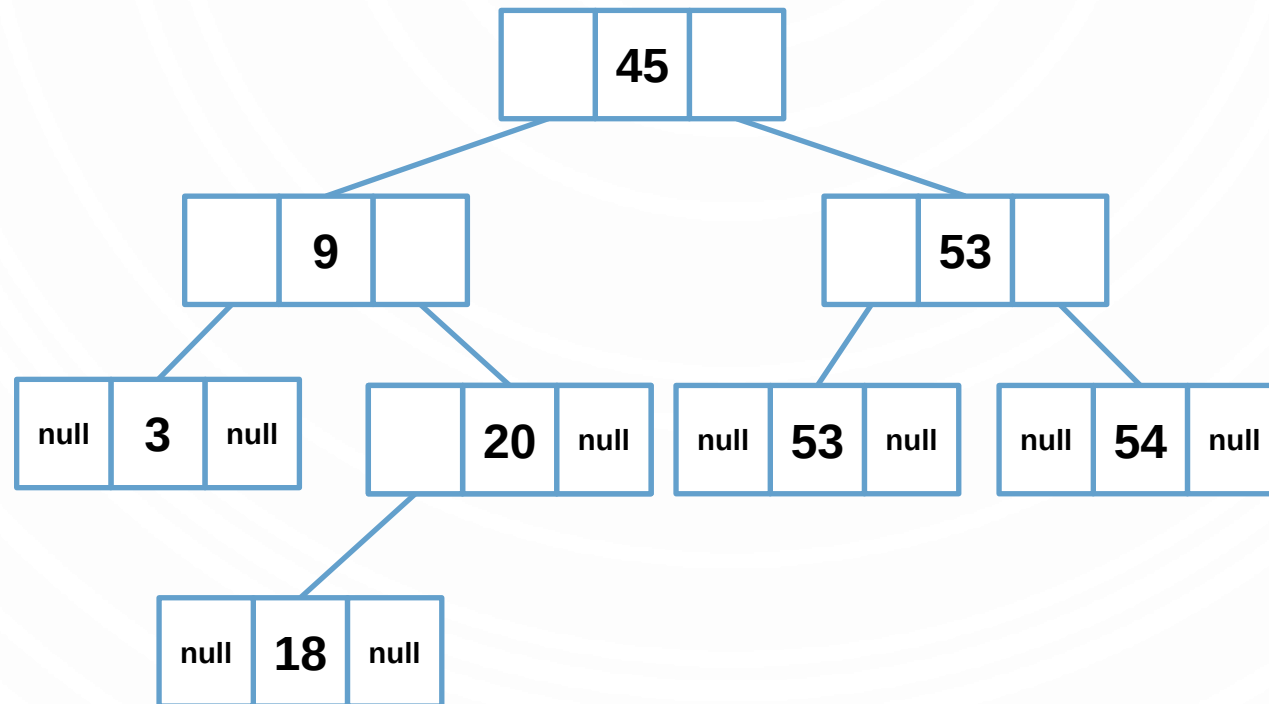
BST – Delete operation

Case 2: deletion of a node with only the right descendant, **17**



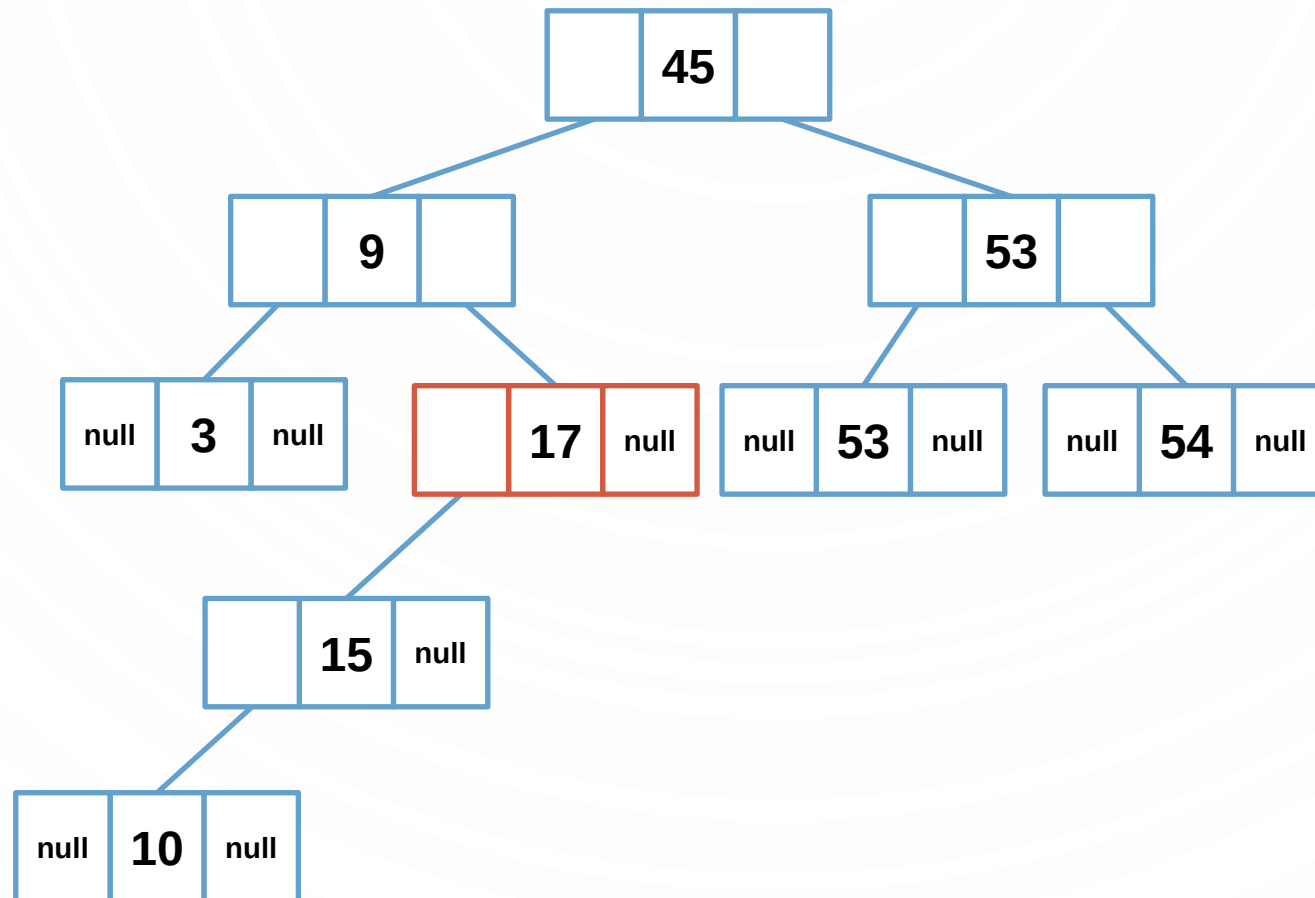
BST – Delete operation

Case 2: deletion of a node with only the right descendant, **17**



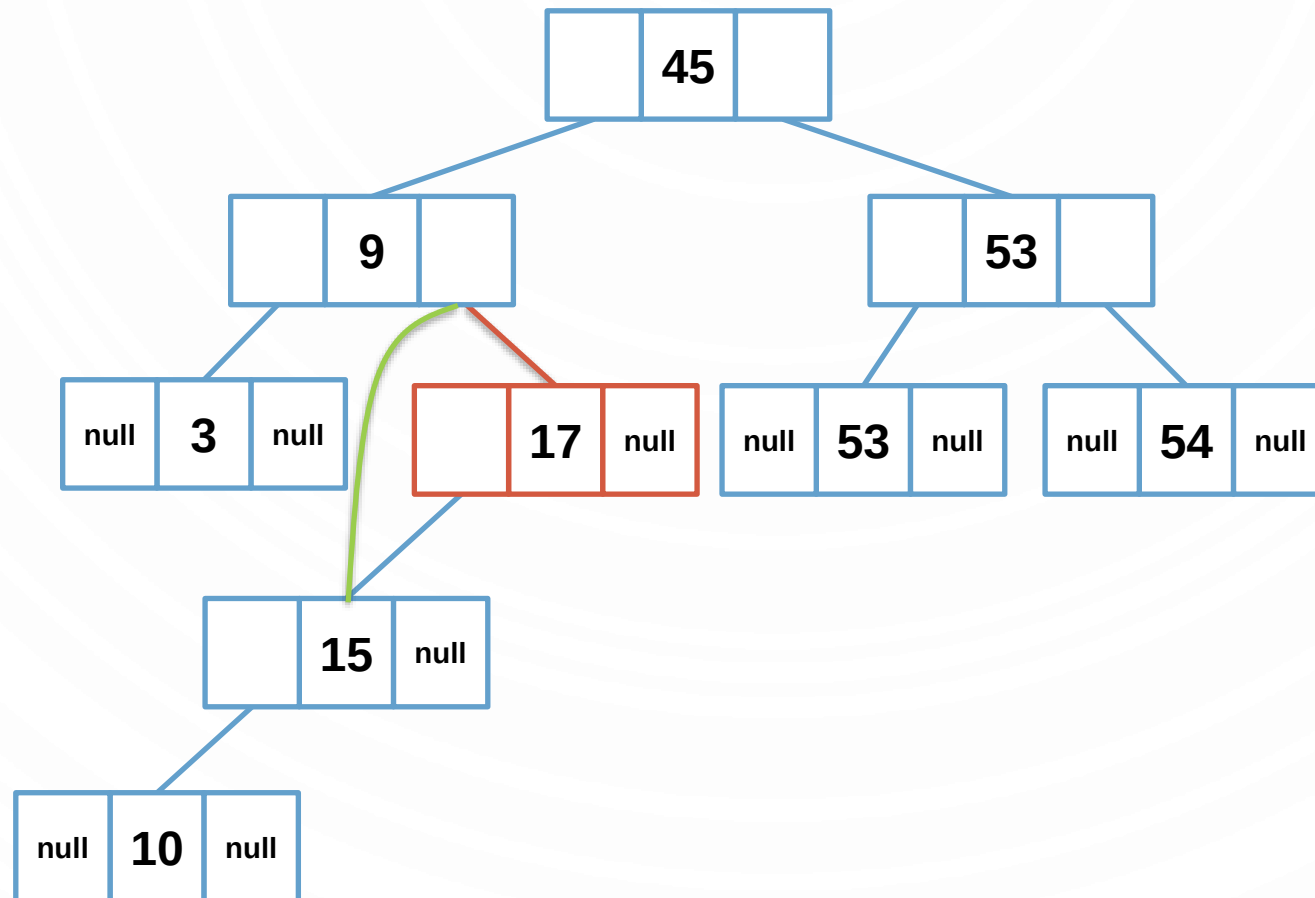
BST – Delete operation

Case 3: deletion of a node with only the left descendant, **17**



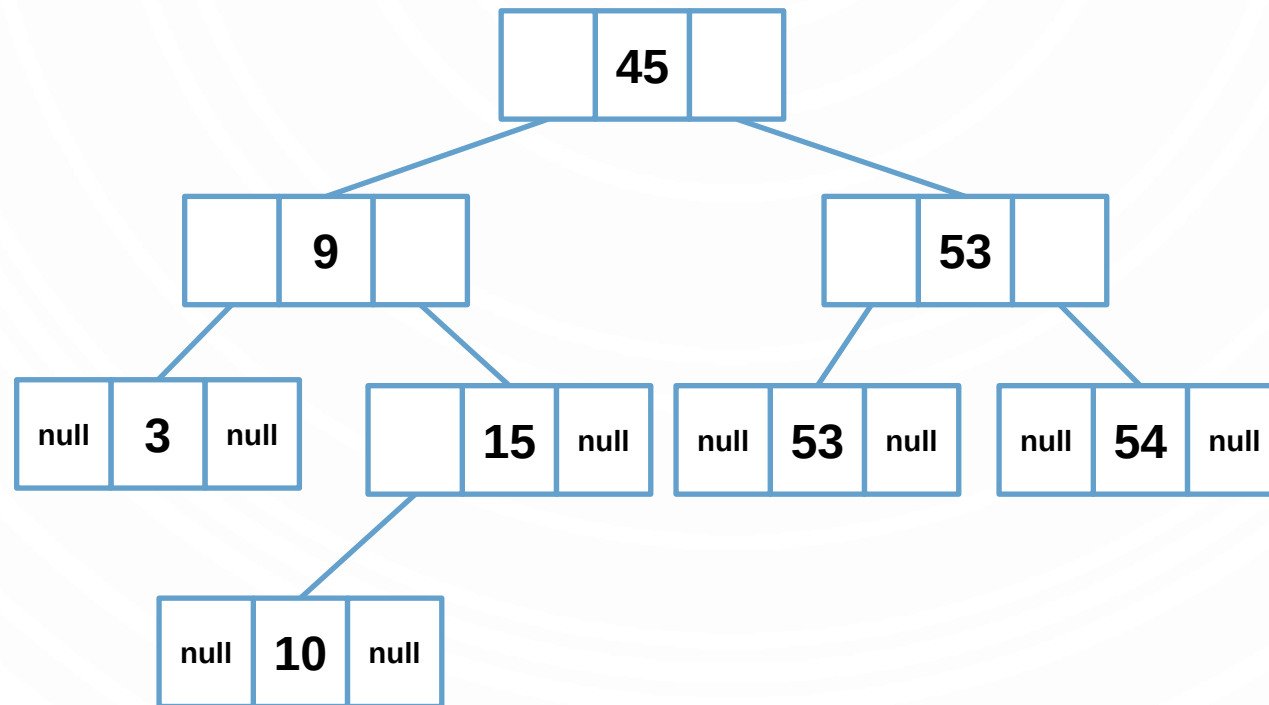
BST – Delete operation

Case 3: deletion of a node with only the left descendant, **17**



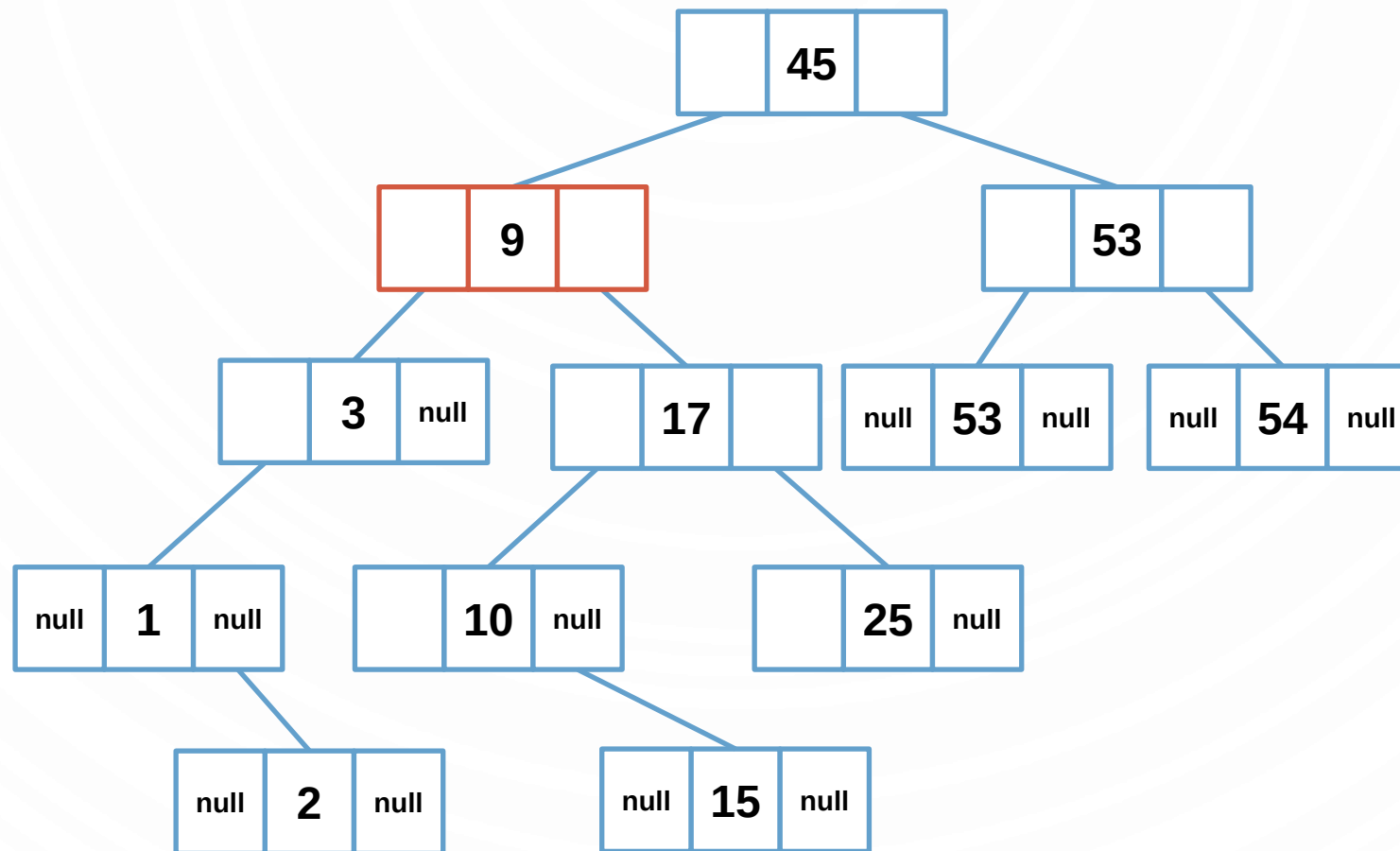
BST – Delete operation

Case 3: deletion of a node with only the left descendant, **17**



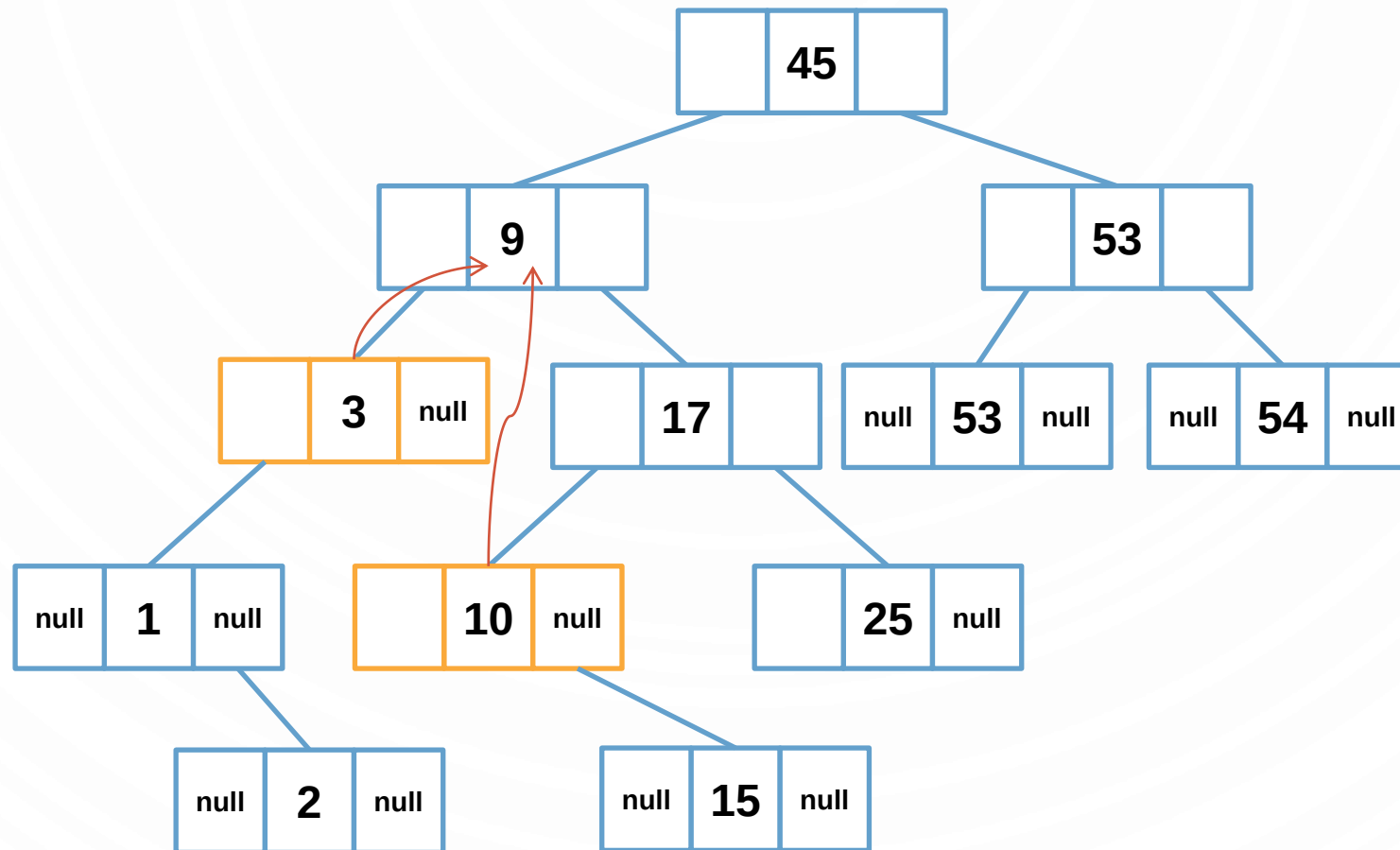
BST – Delete operation

Case 4: deletion of a node with two descendants, **9**



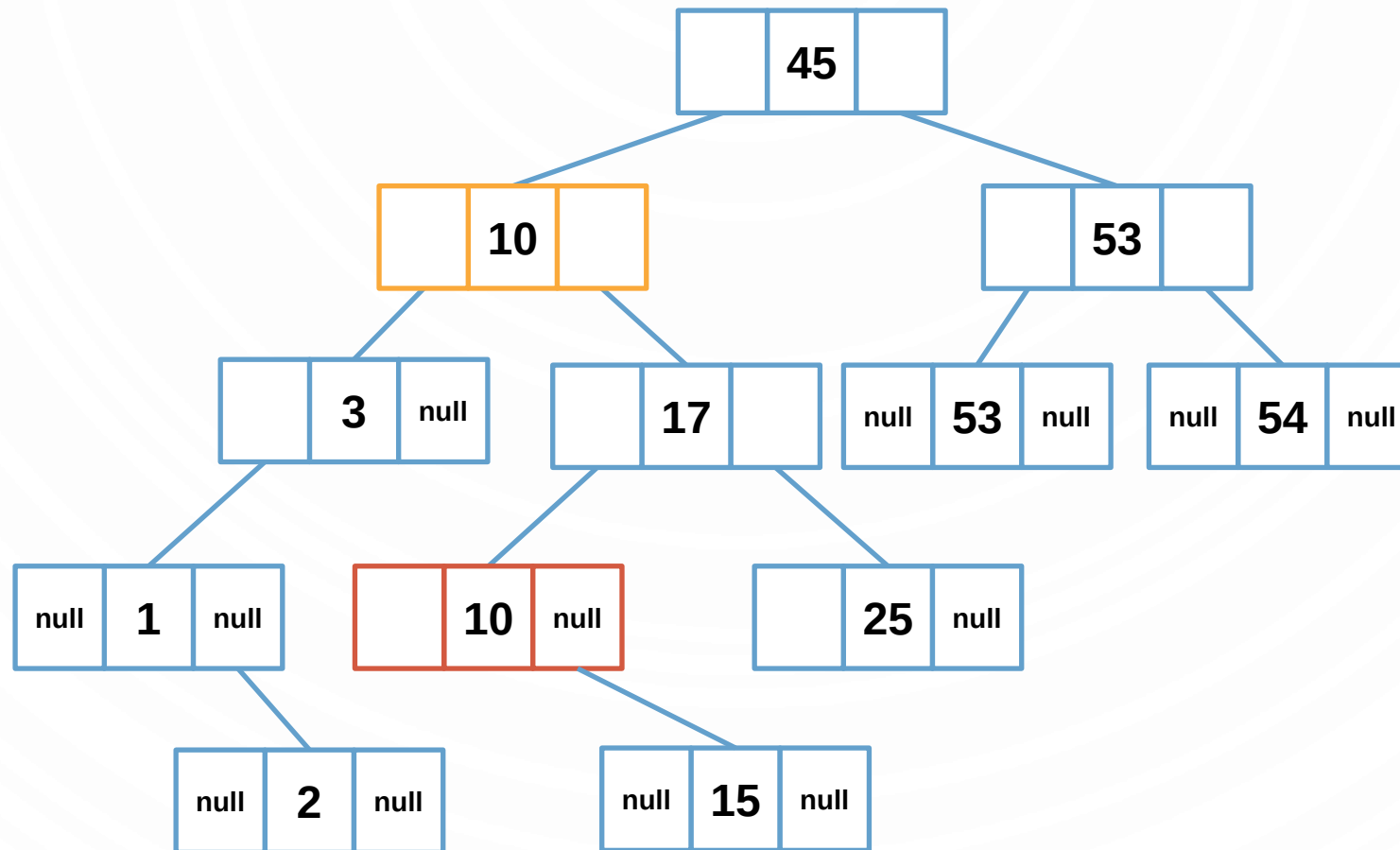
BST – Delete operation

Case 4: deletion of a node with two descendants, **9**



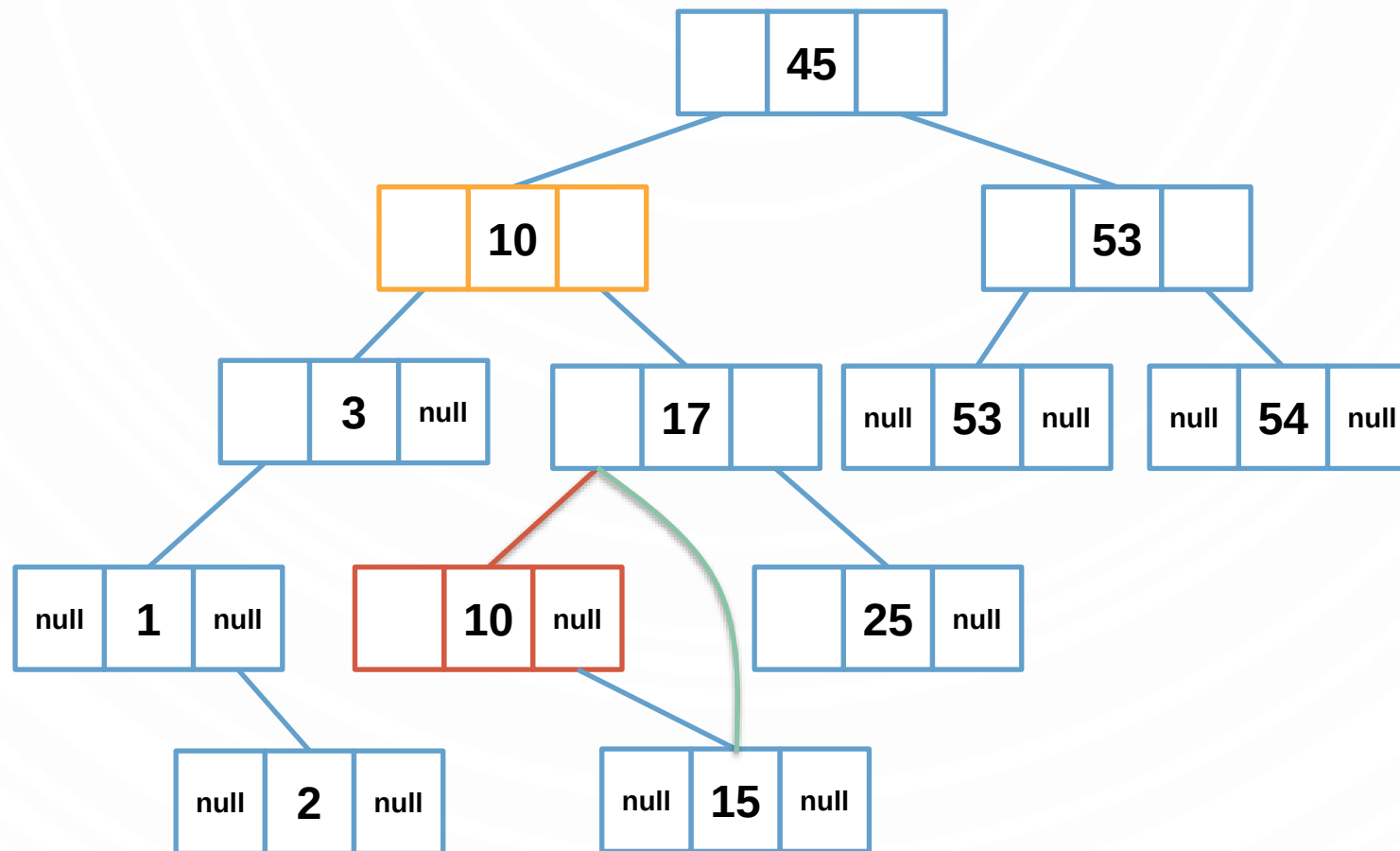
BST – Delete operation

Case 4: deletion of a node with two descendants, **9**



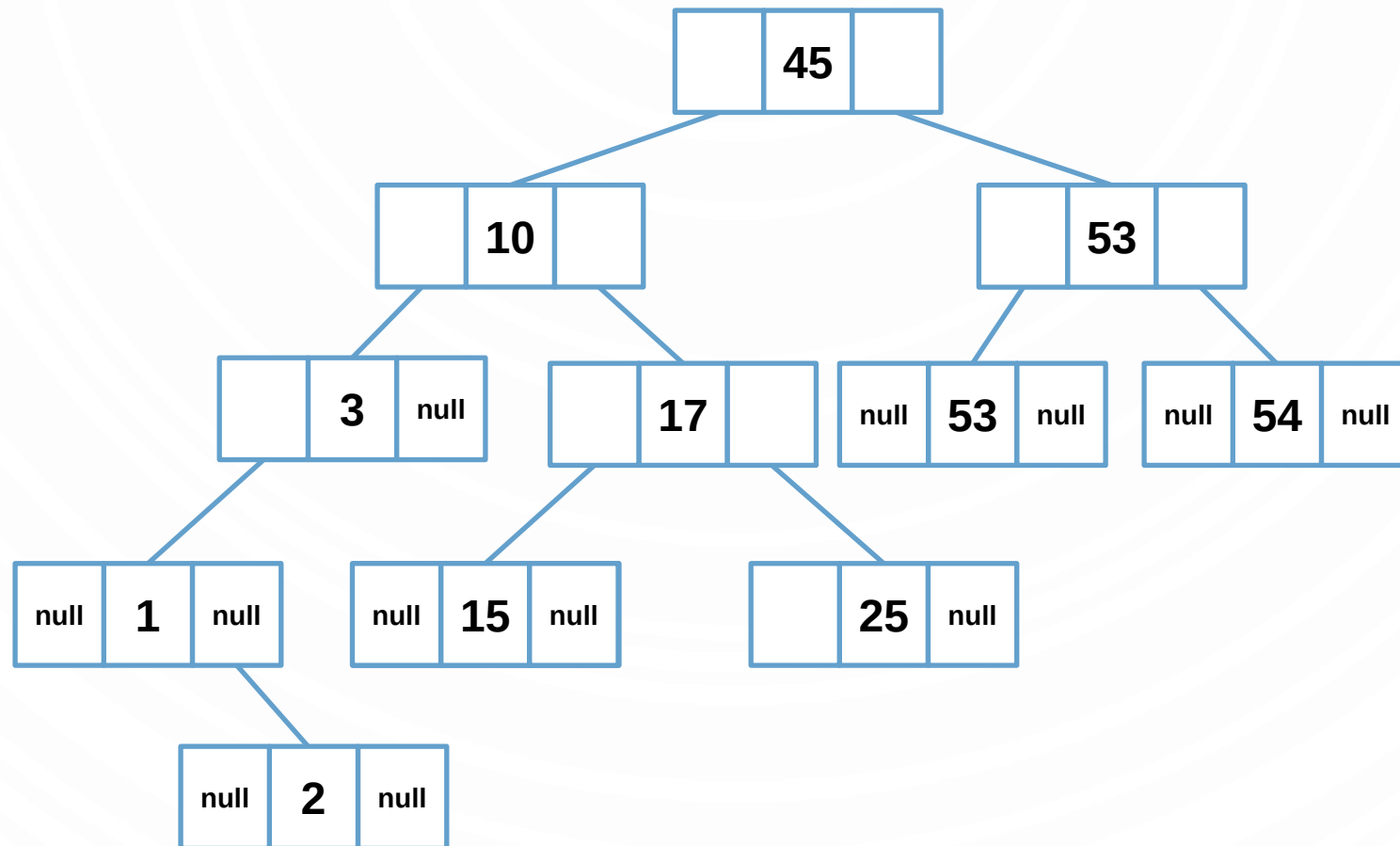
BST – Delete operation

Case 4: deletion of a node with two descendants, **9**



BST – Delete operation

Case 4: deletion of a node with two descendants, **9**



References

- Ion IVAN, Marius POPA, Paul POCATILU, coordonatori – *Structuri de date*, Editura ASE, Bucuresti, 2008, ISBN 978-606-505-031-0
- Ion SMEUREANU – *Programarea in limbajul C/C++*, Editura CISON, 2001
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- <http://www.acs.ase.ro>
- <http://www.itcsolutions.eu>

